Math 510 **Fall 2026**
Study Sheet for Quiz 1 **Drew Armstrong**

Memorize the following definitions and theorems:

- A subset $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ of a vector space V is called *spanning* if for all vectors $\mathbf{v} \in V$ there exist scalars $a_1, \ldots, a_k \in \mathbb{R}$ such that $\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k$.
- A subset $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ of a vector space V is called *independent* if for all scalars $a_1, \ldots, a_k \in \mathbb{R}$ we have
$$a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k = \mathbf{0} \quad \Longrightarrow \quad a_i = 0 \text{ for all } i.$$
- A subset of a vector space is called a *basis* if it is spanning and independent.
- **Generalized Pythagorean Theorem.** The *dot product* in $\mathbb{R}^n$ is defined by
$$\mathbf{x} \bullet \mathbf{y} = (x_1, \ldots, x_n) \bullet (y_1, \ldots, y_n) := x_1y_1 + \cdots + x_ny_n.$$
If $\|\mathbf{x}\|$ is the length of $\mathbf{x}$ then we have $\|\mathbf{x}\|^2 = \mathbf{x} \bullet \mathbf{x}$. More generally, if θ is the angle between the vectors $\mathbf{x}$ and $\mathbf{y}$ then we have
$$\mathbf{x} \bullet \mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta.$$
- Let V be a vector space over $\mathbb{R}$ and consider a function $\mathbf{x}, \mathbf{y} \rightsquigarrow \langle \mathbf{x}, \mathbf{y} \rangle$ that sends ordered pairs of vectors to real numbers. We call this an *inner product* if for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $a, b \in \mathbb{R}$ we have
  - $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$,
  - $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$,
  - $\langle a\mathbf{x}, \mathbf{y} \rangle = a\langle \mathbf{x}, \mathbf{y} \rangle$,
  - $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$,
  - $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ implies $\mathbf{x} = \mathbf{0}$.
- In an inner product space we define the *norm* function by $\|\mathbf{x}\| := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$.
- **Cauchy-Schwarz.** For all vectors $\mathbf{x}, \mathbf{y}$ in an inner product space we have
$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|\|\mathbf{y}\|.$$
- **Steinitz Exchange.** If I is an independent set and S is a spanning set, then $|I| \leq |S|$.
- **Existence of Dimension.** Any two bases for a vector space V have the same number of elements. We define the *dimension*
$$\dim V = \text{the number of elements in any basis.}$$
- Let V be a vector space. A subset $U \subseteq V$ is called a *subspace* if it is "closed under the vector space operations"; that is, if
  - $\mathbf{x}, \mathbf{y} \in U \Rightarrow \mathbf{x} + \mathbf{y} \in U$,
  - $a \in \mathbb{R}, \mathbf{x} \in U \Rightarrow a\mathbf{x} \in U$.